# Group B Quality Assessment

15 assessed report forms | 150 Q1-Q10 item rows | 16 October2026

Use the submitted0-2rubric. Justified N/A removes its maximum. Below60% is sensitivity only. These are first AI assessments; original scores and unperformed independent fields are preserved. ParvaGPU source-version uncertainty and REEF companion identity prevent treating every form as a final study.

| Report | Title | Score / max | Status |
|---|---|---|---|
| GB-P001 | VastPipe | 13/18 | provisionally_eligible |
| GB-P002 | iGniter | 17/20 | provisionally_eligible |
| GB-P003 | gpulets | 18/20 | provisionally_eligible |
| GB-P004 | PipeSwitch | 17/20 | provisionally_eligible |
| GB-P005 | REEF conference | 17/20 | eligible_companion_candidate |
| GB-P006 | Paella: Low-latency Model Serving with Software-defined GPU Scheduling | 16/18 | provisionally_eligible |
| GB-P007 | Orion: Interference-aware, Fine-grained GPU Sharing for ML Applications | 19/20 | provisionally_eligible |
| GB-P008 | GSLICE: Controlled Spatial Sharing of GPUs for a Scalable Inference Platform | 17/20 | provisionally_eligible |
| GB-P009 | Serving DNNs like Clockwork: Performance Predictability from the Bottom Up | 16/20 | provisionally_eligible |
| GB-P010 | Colocating ML Inference and Training with Fast GPU Memory Handover (SIRIUS) | 17/20 | provisionally_eligible_AI_source_assessment |
| GB-P011 | ParvaGPU: Efficient Spatial GPU Sharing for Large-Scale DNN Inference in Cloud Environments | 14/20 | uncertain_publication_version;technical_scope_supported |
| GB-P012 | Torpor: GPU-Enabled Serverless Computing for Low-Latency, Resource-Efficient Inference | 15/20 | provisionally_eligible_AI_source_assessment |
| GB-P013 | StreamBox: A Lightweight GPU SandBox for Serverless Inference Workflow | 15/20 | provisionally_eligible_AI_source_assessment |
| GB-P014 | Efficient Performance-Aware GPU Sharing with Compatibility and Isolation through Kernel Space Interception (KRYPTON) | 16/20 | provisionally_eligible_AI_source_assessment |
| GB-P015 | XSched: Preemptive Scheduling for Diverse XPUs | 16/18 | provisionally_eligible_AI_source_assessment |

# Q1-Q10 checklist

| Item | Criterion | Score2requires |
|---|---|---|
| Q1 | Clarity of objective | Problem, proposed mechanism and evaluated questions are clear. |
| Q2 | Technical description | Sharing primitive, configuration and implementation layer permit classification. |
| Q3 | Experimental transparency | Setup, measurement procedure, repetitions or analytical method and result provenance are clear. |
| Q4 | Hardware/workload description | GPU model/generation, software and concurrent inference workload/tenant characteristics are reported. |
| Q5 | Baseline fairness | Relevant baseline and comparable workload, resources and measurements are documented or justified. |
| Q6 | Utilization/throughput | Claimed benefit has defined metric, unit and method; otherwise alternate outcome is explained. |
| Q7 | Latency measurement | Claimed latency specifies interval, load, statistic and method; claimed tail/SLO results state a quantile. |
| Q8 | Multi-tenant interference | Concurrent tenants/co-tenants are tested and evidence supports interference conclusions. |
| Q9 | Isolation evaluation | Claimed resource, security or fault isolation has a defined boundary and direct test or delimited evidence. |
| Q10 | Replication/limits | Details/artifacts support informed replication; threats, limits or configuration dependence are discussed. |

# GB-P001 | VastPipe

Source: s11390-024-3773-5.md | pp457-459, sections6.7-6.9, Figs14-16

Status: provisionally_eligible. Independent scoring unperformed.

| Q | Score / max | Rationale | Exact source anchor / precision |
|---|---|---|---|
| Q1 | 2/2 | Objective and local GPU portability question explicit | sections1,6.7 \| section_or_figure_only;PDF_page_mapping_pending |
| Q2 | 2/2 | Tuned MPS quotas and CU masking described | sections6.7-6.8 \| section_or_figure_only;PDF_page_mapping_pending |
| Q3 | 1/2 | GPU-specific repeats and variance unresolved in conversion | sections6.7-6.9 \| section_or_figure_only;PDF_page_mapping_pending |
| Q4 | 1/2 | GPU/backend named; precise GPU loads and versions incomplete | sections6.7-6.8 \| section_or_figure_only;PDF_page_mapping_pending |
| Q5 | 1/2 | Same hardware and no/default/tuned MPS; quotas incomplete | Fig14 \| section_or_figure_only;PDF_page_mapping_pending |
| Q6 | 2/2 | Throughput ratios and baseline named in prose | Fig14 and prose \| section_or_figure_only;PDF_page_mapping_pending |
| Q7 | NA/0 | GPU-portability extraction makes no latency claim; do not score NPU latency as GPU evidence | sections6.7-6.8 \| section_or_figure_only;PDF_page_mapping_pending |
| Q8 | 2/2 | Multiple named co-located inference pairs evaluated | Figs14-16 \| section_or_figure_only;PDF_page_mapping_pending |
| Q9 | 1/2 | Performance interference considered; no separate security/fault validation extracted | sections6.7-6.8 \| section_or_figure_only;PDF_page_mapping_pending |
| Q10 | 1/2 | Stable-mix Q-table constraint; GPU replication settings incomplete | section6.9 \| section_or_figure_only;PDF_page_mapping_pending |

Assessor: Codex-GroupB

Original item dates were not populated; no retrospective date invented.

# GB-P002 | iGniter

Source: iGniter user-supplied Markdown | pp820-824, sections4.2,5.1-5.4, Figs13-17

Status: provisionally_eligible. Independent scoring unperformed.

| Q | Score / max | Rationale | Exact source anchor / precision |
|---|---|---|---|
| Q1 | 2/2 | Interference-aware inference provisioning objective explicit | abstract,section1 \| section_or_figure_only;PDF_page_mapping_pending |
| Q2 | 2/2 | MPS allocation and shadow implementation described | section4.2 \| section_or_figure_only;PDF_page_mapping_pending |
| Q3 | 2/2 | Model checks repeated three times with SD; metric windows specified | sections5.1-5.3 \| section_or_figure_only;PDF_page_mapping_pending |
| Q4 | 1/2 | GPU/workloads/backend detailed; OS/driver unresolved | sections4.2,5.1 \| section_or_figure_only;PDF_page_mapping_pending |
| Q5 | 1/2 | Adapted baselines and allocation/batch differences disclosed | section5.1 \| section_or_figure_only;PDF_page_mapping_pending |
| Q6 | 2/2 | Cost, throughput, resource-allocation metrics distinguished | sections5.1,5.3 \| section_or_figure_only;PDF_page_mapping_pending |
| Q7 | 2/2 | P99 windows, SLO and recovery timing explicit | section5.3 Fig17 \| section_or_figure_only;PDF_page_mapping_pending |
| Q8 | 2/2 | Four co-located models and quota configuration specified | section5.2 Fig13 \| section_or_figure_only;PDF_page_mapping_pending |
| Q9 | 1/2 | Performance interference measured; security/fault boundaries not tested in checked portion | sections5.2-5.3 \| section_or_figure_only;PDF_page_mapping_pending |
| Q10 | 2/2 | Code, prediction limitations and PCIe caveat explicit | sections4.2,5.2 \| section_or_figure_only;PDF_page_mapping_pending |

Assessor: Codex-GroupB

Original item dates were not populated; no retrospective date invented.

# GB-P003 | gpulets

Source: https://www.usenix.org/system/files/atc22-choi-seungbeom.pdf | pp201-203 local mechanism/Fig7; p207 Tables2-3;pp208-209 system methodology

Status: provisionally_eligible. Independent scoring unperformed.

| Q | Score / max | Rationale | Exact source anchor / precision |
|---|---|---|---|
| Q1 | 2/2 | Clear SLO-aware co-location objective | p199 abstract;section1 \| printed_page_and_section_or_figure |
| Q2 | 2/2 | MPS partitions and temporal batching described | pp201-207 sections2-4 \| printed_page_and_section_or_figure |
| Q3 | 2/2 | Poisson requests;median of3 repeats;throughput/SLO procedure | pp207-208 section5.1-5.2 \| printed_page_and_section_or_figure |
| Q4 | 2/2 | Full GPU/CPU/OS/driver/runtime and model tables | p207 Tables2-3 \| printed_page_and_section_or_figure |
| Q5 | 2/2 | Ported SBP comparator and Nexus incompatibility justified | p207 section5.1 \| printed_page_and_section_or_figure |
| Q6 | 2/2 | Maximum throughput with99%SLO definition;system scale explicit | p208 section5.2 \| printed_page_and_section_or_figure |
| Q7 | 2/2 | SLO99% procedure;local overhead CDF population explicit | p203 Fig7;p208 section5.2 \| printed_page_and_section_or_figure |
| Q8 | 2/2 | Paired model interference and scheduler interference comparison | p203 Fig7;p208 Fig14 \| printed_page_and_section_or_figure |
| Q9 | 1/2 | Performance interference checked;security/fault boundary not extracted | p203 Fig7;section5.2 \| printed_page_and_section_or_figure |
| Q10 | 1/2 | Prototype and appendix setup available;replication not attempted | p207 Table2;appendixA \| printed_page_and_section_or_figure |

Assessor: Codex-GroupB

Original dates: 2026-10-15T22:10:27+00:00

# GB-P004 | PipeSwitch

Source: https://www.usenix.org/system/files/osdi20-bai.pdf | p508 section6/6.1; p509 Tables3-4

Status: provisionally_eligible. Independent scoring unperformed.

| Q | Score / max | Rationale | Exact source anchor / precision |
|---|---|---|---|
| Q1 | 2/2 | Fine-grained task-switching problem and design explicit | p499 abstract;section1 \| printed_page_and_section_or_figure |
| Q2 | 2/2 | Controller,memory daemon,workers and pipeline described | pp504-508 sections4-5 \| printed_page_and_section_or_figure |
| Q3 | 2/2 | 100-run mean;min/max forFig6;setup described | p508 section6 \| printed_page_and_section_or_figure |
| Q4 | 1/2 | Hardware/software specified;OS/kernel/driver not extracted | p508 section6 \| printed_page_and_section_or_figure |
| Q5 | 2/2 | Same modified PyTorch;ready-model and swapping conditions explicit | p508 section6.1 \| printed_page_and_section_or_figure |
| Q6 | 2/2 | Inference throughput and utilization denominator defined | p509 section6.1 Fig6 \| printed_page_and_section_or_figure |
| Q7 | 2/2 | End-to-end overhead defined;mean and extrema identified | pp508-509 Table3 \| printed_page_and_section_or_figure |
| Q8 | 2/2 | Training/inference co-location;inference metric separated | pp508-509 section6.1 \| printed_page_and_section_or_figure |
| Q9 | 1/2 | Process/memory control described;security/fault testing not extracted | pp507-508 implementation \| printed_page_and_section_or_figure |
| Q10 | 1/2 | Versioned stack described;limitations documented;replication not attempted | p508 section6;section7 \| printed_page_and_section_or_figure |

Assessor: Codex-GroupB

Original dates: 2026-10-15T22:10:27+00:00

# GB-P005 | REEF conference

Source: https://www.usenix.org/system/files/osdi22-han.pdf | p548 section7.1 Table2;p549 section7.2 Fig11;section8 idempotence limitation

Status: eligible_companion_candidate. Independent scoring unperformed.

| Q | Score / max | Rationale | Exact source anchor / precision |
|---|---|---|---|
| Q1 | 2/2 | Concurrent real-time/best-effort inference objective explicit | p539 abstract;section1 \| printed_page_and_section_or_figure |
| Q2 | 2/2 | Reset preemption,queues and padding implementation described | sections3-6;pp547-548 \| printed_page_and_section_or_figure |
| Q3 | 1/2 | Benchmark arrivals and baseline methods detailed;repeat/variance protocol not extracted | p548 section7.1 \| printed_page_and_section_or_figure |
| Q4 | 2/2 | CPU,GPU,host RAM,OS,runtime,models andclients specified | p548 section7.1 Table2 \| printed_page_and_section_or_figure |
| Q5 | 2/2 | RT-only and GPUStreams/SEQ conditions;extensions disclosed | p548 section7.1;p549 section7.3 \| printed_page_and_section_or_figure |
| Q6 | 2/2 | Total RT+BE throughput and denominator explicit | p549 section7.2 Fig11 \| printed_page_and_section_or_figure |
| Q7 | 1/2 | End-to-end latency and load specified;request percentile not extracted | pp548-549 Table2/Fig11 \| printed_page_and_section_or_figure |
| Q8 | 2/2 | MultipleRT/BE clients and multiple mixes directly tested | p548 Table2;p549 section7.2 \| printed_page_and_section_or_figure |
| Q9 | 1/2 | Performance isolation tested;no extracted memory/security/fault boundary test | pp548-549 section7.2 \| printed_page_and_section_or_figure |
| Q10 | 2/2 | Artifact and idempotence/platform limitations explicit | p540 repository;section8;appendixA \| printed_page_and_section_or_figure |

Assessor: Codex-GroupB

Original dates: 2026-10-15T22:10:27+00:00

# GB-P006 | Paella: Low-latency Model Serving with Software-defined GPU Scheduling

Source: https://kelvin-ng.github.io/assets/sosp2023-final224.pdf | printed pp604-606; PDF pp10-12; sections7.1-7.3

Status: provisionally_eligible;study_identity_provisional. Independent scoring unperformed.

| Q | Score / max | Rationale | Exact source anchor / precision |
|---|---|---|---|
| Q1 | 2/2 | Explicit goal | p595 section1 \| printed_page_and_section_or_figure |
| Q2 | 2/2 | Mechanism classifiable | pp599-603 sections4-6 \| printed_page_and_section_or_figure |
| Q3 | 1/2 | Repetition count unclear | p604 methodology \| printed_page_and_section_or_figure |
| Q4 | 2/2 | Versioned setup | p604 methodology \| printed_page_and_section_or_figure |
| Q5 | 1/2 | Wrapper caveat | p604 Table3/baselines \| printed_page_and_section_or_figure |
| Q6 | 2/2 | Defined throughput | pp604-605 Figs11-12 \| printed_page_and_section_or_figure |
| Q7 | 2/2 | p99 specified | pp604-605 Figs11-12 \| printed_page_and_section_or_figure |
| Q8 | 2/2 | Mixed jobs tested | p605 Figs12-13 \| printed_page_and_section_or_figure |
| Q9 | NA/0 | No security/fault or resource-isolation guarantee appraised; fairness is assessed under Q8; N/A provisional | pp606-607 discussion;scope provisional \| printed_page_and_section_or_figure |
| Q10 | 2/2 | Artifact and limits | pp606-607 section8 \| printed_page_and_section_or_figure |

Assessor: Codex-GroupB

Original dates: 2026-10-16T10:00:46+00:00

# GB-P007 | Orion: Interference-aware, Fine-grained GPU Sharing for ML Applications

Source: https://anakli.inf.ethz.ch/papers/orion_eurosys24.pdf | printed pp1082-1087; PDF pp8-13; sections6-7

Status: provisionally_eligible;study_identity_provisional. Independent scoring unperformed.

| Q | Score / max | Rationale | Exact source anchor / precision |
|---|---|---|---|
| Q1 | 2/2 | Explicit goal | p1075 section1 \| printed_page_and_section_or_figure |
| Q2 | 2/2 | Interception specified | pp1079-1082 sections4-5 \| printed_page_and_section_or_figure |
| Q3 | 2/2 | Three repetitions | p1082 section6.1 \| printed_page_and_section_or_figure |
| Q4 | 2/2 | Versioned setup | p1082 section6.1 \| printed_page_and_section_or_figure |
| Q5 | 1/2 | Reimplemented comparator | p1083 baselines \| printed_page_and_section_or_figure |
| Q6 | 2/2 | Defined aggregate metric | p1086 section6.2.3 \| printed_page_and_section_or_figure |
| Q7 | 2/2 | p99 and arrivals | p1086 section6.2.3 \| printed_page_and_section_or_figure |
| Q8 | 2/2 | Collocations tested | p1086 section6.3 \| printed_page_and_section_or_figure |
| Q9 | 2/2 | Trust boundary explicit | p1087 section7 Security \| printed_page_and_section_or_figure |
| Q10 | 2/2 | Artifact and limitations | p1087 section7;artifact appendix \| printed_page_and_section_or_figure |

Assessor: Codex-GroupB

Original dates: 2026-10-16T10:00:46+00:00

# GB-P008 | GSLICE: Controlled Spatial Sharing of GPUs for a Scalable Inference Platform

Source: https://par.nsf.gov/servlets/purl/10299299 | printed pp499-501; PDF pp8-10; sections4.1-4.3

Status: provisionally_eligible;study_identity_provisional. Independent scoring unperformed.

| Q | Score / max | Rationale | Exact source anchor / precision |
|---|---|---|---|
| Q1 | 2/2 | Explicit goal | p492 introduction \| printed_page_and_section_or_figure |
| Q2 | 2/2 | Sharing described | pp495-499 section3 \| printed_page_and_section_or_figure |
| Q3 | 1/2 | Confidence intervals; counts unclear | p500 error bars;section4 \| printed_page_and_section_or_figure |
| Q4 | 2/2 | Hardware and OS | pp499-500 testbed \| printed_page_and_section_or_figure |
| Q5 | 2/2 | Comparator scope stated | p501 section4.3 \| printed_page_and_section_or_figure |
| Q6 | 2/2 | GUE defined | p494 GUE definition;p500 Table3 \| printed_page_and_section_or_figure |
| Q7 | 1/2 | Tail statistic unclear | pp500-501 latency plots \| printed_page_and_section_or_figure |
| Q8 | 2/2 | Concurrent functions | p500 section4.2 \| printed_page_and_section_or_figure |
| Q9 | 2/2 | Fixed quotas and performance-isolation tests; no stronger security claim inferred | pp496-498 resource controls;p502 performance isolation \| printed_page_and_section_or_figure |
| Q10 | 1/2 | Reproduction incomplete | pp499-504 implementation/evaluation/limits \| printed_page_and_section_or_figure |

Assessor: Codex-GroupB

Original dates: 2026-10-16T10:02:57+00:00

# GB-P009 | Serving DNNs like Clockwork: Performance Predictability from the Bottom Up

Source: https://people.mpi-sws.org/~antoinek/documents/20osdi_clockwork.pdf | printed pp451-452; author PDF pp9-10; section6.1/Fig5

Status: provisionally_eligible;study_identity_provisional. Independent scoring unperformed.

| Q | Score / max | Rationale | Exact source anchor / precision |
|---|---|---|---|
| Q1 | 2/2 | Explicit goal | p443 introduction \| printed_page_and_section_or_figure |
| Q2 | 2/2 | Scheduling described | pp448-451 sections4-5 \| printed_page_and_section_or_figure |
| Q3 | 1/2 | Repetition count unclear | p451 section6 setup \| printed_page_and_section_or_figure |
| Q4 | 1/2 | Versions not extracted | p451 section6 setup \| printed_page_and_section_or_figure |
| Q5 | 1/2 | Hardware mismatch disclosed | p451 section6.1 mismatch \| printed_page_and_section_or_figure |
| Q6 | 2/2 | Goodput defined | p452 goodput definition \| printed_page_and_section_or_figure |
| Q7 | 2/2 | SLO and CDF | pp451-452 Fig5 \| printed_page_and_section_or_figure |
| Q8 | 2/2 | Contending models | p451 section6.1 \| printed_page_and_section_or_figure |
| Q9 | 1/2 | No fault guarantee | pp455-456 discussion \| printed_page_and_section_or_figure |
| Q10 | 2/2 | Artifact and limits | pp455-456 discussion;artifact appendix \| printed_page_and_section_or_figure |

Assessor: Codex-GroupB

Original dates: 2026-10-16T10:02:57+00:00

# GB-P010 | Colocating ML Inference and Training with Fast GPU Memory Handover (SIRIUS)

Source: https://www.usenix.org/system/files/atc25-wang-jiali.pdf | printed pp1665-1666, section6/6.1, Table3, Figure10; PDF pp10-11 (includes cover)

Status: provisionally_eligible_AI_source_assessment. Independent scoring unperformed.

| Q | Score / max | Rationale | Exact source anchor / precision |
|---|---|---|---|
| Q1 | 2/2 | Inference/training colocation objective and scope explicit. | p1657 section1 \| printed_page_and_section_or_figure |
| Q2 | 2/2 | VMM pool, memory handover and SM-mask control described. | pp1660-1665 sections3-5 \| printed_page_and_section_or_figure |
| Q3 | 1/2 | Trace distributions and300s windows described; repeated-run uncertainty not established in checked evaluation. | pp1665-1666 section6 \| printed_page_and_section_or_figure |
| Q4 | 2/2 | GPU, CPU/RAM, software and model sizes specified. | p1665 testbed; p1666 Table3 \| printed_page_and_section_or_figure |
| Q5 | 1/2 | TaskSwitch is an adapted implementation; baseline choices explicit but implementation equivalence remains limited. | p1665 comparing targets \| printed_page_and_section_or_figure |
| Q6 | 2/2 | Training samples/s is defined; inference latency/SLO are kept distinct. | p1666 metrics/Figure10 \| printed_page_and_section_or_figure |
| Q7 | 2/2 | P99, request arrivals, window and SLO definition supplied. | p1666 section6.1/Figure10 \| printed_page_and_section_or_figure |
| Q8 | 2/2 | Co-located inference/training traces tested on one GPU. | p1666 section6.1/Figure10 \| printed_page_and_section_or_figure |
| Q9 | 1/2 | Memory-reclamation boundary described; stronger fault/security validation not established. | pp1661-1665 sections4-5 \| printed_page_and_section_or_figure |
| Q10 | 2/2 | Detailed implementation and evaluated artifact supplied; scope limitations discussed. | p1665 section5; p1669 section7; artifact appendix \| printed_page_and_section_or_figure |

Assessor: Codex-GroupB source assessment

Original dates: 2026-10-16T12:35:50+00:00

# GB-P011 | ParvaGPU: Efficient Spatial GPU Sharing for Large-Scale DNN Inference in Cloud Environments

Source: https://arxiv.org/pdf/2409.14447 | author preprint PDF pp4-5 Figures3-4; p8 experimental environment

Status: uncertain_publication_version;technical_scope_supported. Independent scoring unperformed.

| Q | Score / max | Rationale | Exact source anchor / precision |
|---|---|---|---|
| Q1 | 2/2 | Clear inference resource-sharing objective. | PDF pp1-2 introduction \| printed_page_and_section_or_figure |
| Q2 | 2/2 | MIG/MPS segments and algorithms explained. | PDF pp4-7 sectionIII \| printed_page_and_section_or_figure |
| Q3 | 1/2 | Profiling procedure and fleet scenarios described; repetition/uncertainty reporting unclear. | PDF pp5,8 sectionIV \| printed_page_and_section_or_figure |
| Q4 | 1/2 | Hardware/workload described; runtime version1.14.0 is printed but needs verification. | PDF p8 sectionIV-A \| printed_page_and_section_or_figure |
| Q5 | 1/2 | Local process comparison interpretable; fleet baselines adapted and no final-version equivalence established. | PDF pp5,8 \| printed_page_and_section_or_figure |
| Q6 | 2/2 | Throughput axes specify requests/s and exact local values are given. | PDF pp4-5 Figure3 \| printed_page_and_section_or_figure |
| Q7 | 1/2 | Latency milliseconds and settings stated; profiler quantile unclear. | PDF pp4-5 Figure4 \| printed_page_and_section_or_figure |
| Q8 | 2/2 | Concurrent MPS process counts evaluated inside one MIG segment. | PDF pp4-5 Figures3-4 \| printed_page_and_section_or_figure |
| Q9 | 1/2 | MIG boundary described; no direct security/fault validation established. | PDF pp2-4 background/design \| printed_page_and_section_or_figure |
| Q10 | 1/2 | Algorithms/settings detailed; accessed version remains preprint and replication verification incomplete. | PDF pp5-10 \| printed_page_and_section_or_figure |

Assessor: Codex-GroupB source assessment

Original dates: 2026-10-16T12:35:50+00:00

# GB-P012 | Torpor: GPU-Enabled Serverless Computing for Low-Latency, Resource-Efficient Inference

Source: https://www.usenix.org/system/files/atc25-yu.pdf | pp604-606 sections7.1-7.2, Table4, Figure8; PDF pp9-11

Status: provisionally_eligible_AI_source_assessment. Independent scoring unperformed.

| Q | Score / max | Rationale | Exact source anchor / precision |
|---|---|---|---|
| Q1 | 2/2 | Inference memory/density objective explicit. | p597 section1 \| printed_page_and_section_or_figure |
| Q2 | 2/2 | API redirection, pooling and scheduling described. | pp600-604 sections4-6 \| printed_page_and_section_or_figure |
| Q3 | 1/2 | Production trace and request rates given; repeated-run uncertainty not established. | pp604-606 section7 \| printed_page_and_section_or_figure |
| Q4 | 1/2 | GPU/CPU/RAM supplied; exact software version matrix missing in checked setup. | p604 settings \| printed_page_and_section_or_figure |
| Q5 | 1/2 | Native and adapted INFless-KA comparison described; resource/CPU paths differ. | pp605-606 sections7.1-7.3 \| printed_page_and_section_or_figure |
| Q6 | 2/2 | Throughput/load definitions and local normalization supplied. | pp605-606 Figure8 \| printed_page_and_section_or_figure |
| Q7 | 2/2 | P98 and per-function load/SLO explicit. | pp605-606 metrics/Figure8 \| printed_page_and_section_or_figure |
| Q8 | 2/2 | Multiple functions share a single GPU in Figure8. | pp605-606 Figure8 \| printed_page_and_section_or_figure |
| Q9 | 1/2 | Sharing/isolation modes separated; strong security/fault tests not established for evaluated mode. | sections4,7-8 \| section_or_figure_only;PDF_page_mapping_pending |
| Q10 | 1/2 | Implementation detail and limitations supplied; independent artifact reproduction unperformed. | sections6-8 \| section_or_figure_only;PDF_page_mapping_pending |

Assessor: Codex-GroupB source assessment

Original dates: 2026-10-16T12:41:06+00:00

# GB-P013 | StreamBox: A Lightweight GPU SandBox for Serverless Inference Workflow

Source: https://www.usenix.org/system/files/atc24-wu-hao.pdf | pp67-70 sections9.1-9.4; Figures12-14,19; PDF pp10-13

Status: provisionally_eligible_AI_source_assessment. Independent scoring unperformed.

| Q | Score / max | Rationale | Exact source anchor / precision |
|---|---|---|---|
| Q1 | 2/2 | Workflow-local runtime sharing objective explicit. | p59 section1 \| printed_page_and_section_or_figure |
| Q2 | 2/2 | Streams,PTB compiler and memory management explained. | pp62-67 sections3-8 \| printed_page_and_section_or_figure |
| Q3 | 1/2 | Production-style traces and ablations provided; repeated-run uncertainty unclear. | pp67-70 section9 \| printed_page_and_section_or_figure |
| Q4 | 1/2 | Hardware/frameworks named; OS/driver and version compatibility need checking. | p67 setup \| printed_page_and_section_or_figure |
| Q5 | 1/2 | Common optimizations enabled, but baselines adapted and batch choices affect ratios. | p67 Table2/comparing targets \| printed_page_and_section_or_figure |
| Q6 | 2/2 | Memory and throughput units/ratios defined. | p68 Figures12-13 \| printed_page_and_section_or_figure |
| Q7 | 1/2 | CDF/SLO reported; no precise tail scalar extracted. | p68 Figure14 \| printed_page_and_section_or_figure |
| Q8 | 2/2 | Explicit3/6-user single-GPU test. | p70 Figure19 \| printed_page_and_section_or_figure |
| Q9 | 1/2 | Same-runtime sharing weakens function isolation; security boundary is deliberately scoped. | pp62-63; p70 related-work boundary \| printed_page_and_section_or_figure |
| Q10 | 2/2 | Detailed implementation, preparation overhead and multi-user limits supplied. | p70 sections9.3-9.4 \| printed_page_and_section_or_figure |

Assessor: Codex-GroupB source assessment

Original dates: 2026-10-16T12:41:06+00:00

# GB-P014 | Efficient Performance-Aware GPU Sharing with Compatibility and Isolation through Kernel Space Interception (KRYPTON)

Source: https://www.usenix.org/system/files/atc25-zhang-shulai.pdf | pp1011-1015 sections5.1,5.5-5.7,7; PDF pp10-14

Status: provisionally_eligible_AI_source_assessment. Independent scoring unperformed.

| Q | Score / max | Rationale | Exact source anchor / precision |
|---|---|---|---|
| Q1 | 2/2 | Sharing/isolation/compatibility objectives explicit. | p1003 section1 \| printed_page_and_section_or_figure |
| Q2 | 2/2 | Kernel interception,token control and MIG orchestration described. | pp1006-1011 sections3-4 \| printed_page_and_section_or_figure |
| Q3 | 1/2 | Setup and workload matrix supplied; repeated-run uncertainty unclear. | p1011 section5.1 \| printed_page_and_section_or_figure |
| Q4 | 2/2 | Hardware,OS,kernel,driver and frameworks specified. | p1011 Tables1-2 \| printed_page_and_section_or_figure |
| Q5 | 1/2 | Isolation-capable baselines and adapters disclosed; equivalence limited. | pp1011-1012 comparing targets \| printed_page_and_section_or_figure |
| Q6 | 2/2 | Quota/throughput metrics explicitly normalized. | pp1013-1014 sections5.4-5.6 \| printed_page_and_section_or_figure |
| Q7 | 2/2 | QoS uses originalP99 and defined Poisson load/token lengths. | p1014 Figure17/section5.5 \| printed_page_and_section_or_figure |
| Q8 | 2/2 | Local co-running inference error measured. | p1014 section5.6 \| printed_page_and_section_or_figure |
| Q9 | 1/2 | Boundary/security rationale described; direct fault/security validation not established in checked evaluation. | p1015 section7 \| printed_page_and_section_or_figure |
| Q10 | 1/2 | Implementation andMIGlimitations supplied; independent artifact reproduction unperformed. | pp1011,1015 sections4,7 \| printed_page_and_section_or_figure |

Assessor: Codex-GroupB source assessment

Original dates: 2026-10-16T12:41:06+00:00

# GB-P015 | XSched: Preemptive Scheduling for Diverse XPUs

Source: https://www.usenix.org/system/files/osdi25-shen-weihang.pdf | p683 section7.4 case3/Figure15; PDF p14

Status: provisionally_eligible_AI_source_assessment. Independent scoring unperformed.

| Q | Score / max | Rationale | Exact source anchor / precision |
|---|---|---|---|
| Q1 | 2/2 | Portable scheduling objective andGPUcase clear. | pp671-672 section1 \| printed_page_and_section_or_figure |
| Q2 | 2/2 | XQueue abstraction andmulti-level preemption described. | pp674-679 sections3-6 \| printed_page_and_section_or_figure |
| Q3 | 1/2 | Workload rates and policies explicit; repeat uncertainty not established. | pp680-683 section7 \| printed_page_and_section_or_figure |
| Q4 | 1/2 | GPU/model/rates explicit;host/software versions incomplete in extracted case. | p683 case3 \| printed_page_and_section_or_figure |
| Q5 | 2/2 | Same-GPU standalone/native priority comparisons supplied. | p683 Figure15 \| printed_page_and_section_or_figure |
| Q6 | 2/2 | Throughput-latency curves defined forGPUinferencecase. | p683 Figure15b \| printed_page_and_section_or_figure |
| Q7 | 2/2 | P99 and offered rates explicit. | p683 Figure15a \| printed_page_and_section_or_figure |
| Q8 | 2/2 | Two concurrentBertclients ononeGV100. | p683 case3 \| printed_page_and_section_or_figure |
| Q9 | NA/0 | No claimed security/fault-containment guarantee in extracted scheduling claim;N/A provisional for claim-specificQ9. | pp671-672 scope; p683 case3 \| printed_page_and_section_or_figure |
| Q10 | 2/2 | Implementation/interfaces,artifact andhardware-dependence detailed. | sections6-8;artifact appendix \| section_or_figure_only;PDF_page_mapping_pending |

Assessor: Codex-GroupB source assessment

Original dates: 2026-10-16T12:41:06+00:00